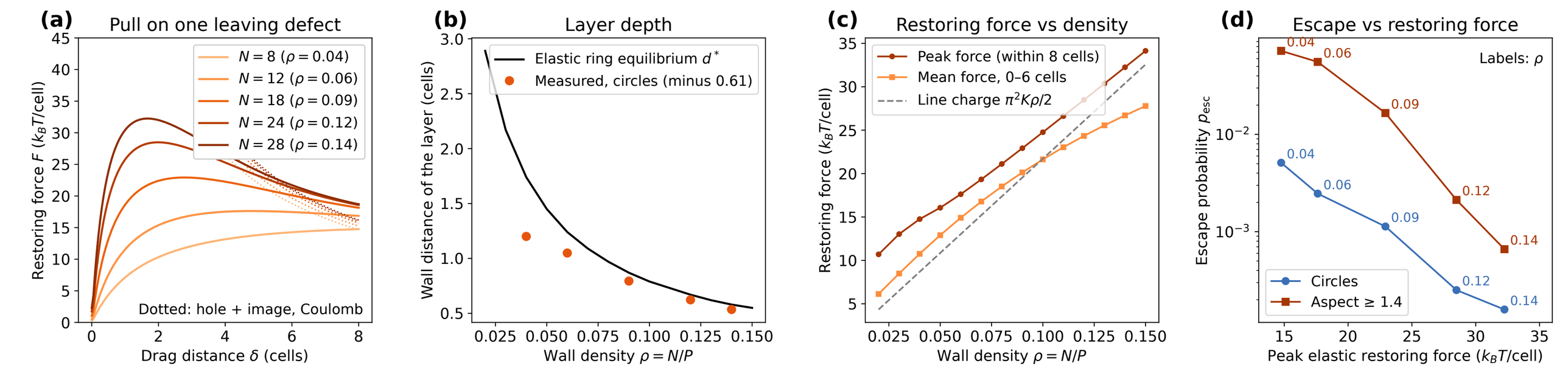
(a)
Pull on one leaving defect
Restoring force F (k_BT/cell)
Drag distance δ (cells)
$N = 8$ ($\rho = 0.04$)
$N = 12$ ($\rho = 0.06$)
$N = 18$ ($\rho = 0.09$)
$N = 24$ ($\rho = 0.12$)
$N = 28$ ($\rho = 0.14$)
Dotted: hole + image, Coulomb
(b)
Layer depth
Wall distance of the layer (cells)
Wall density $\rho = N/P$
Elastic ring equilibrium d^*
Measured, circles (minus 0.61)
(c)
Restoring force vs density
Restoring force (k_BT/cell)
Wall density $\rho = N/P$
Peak force (within 8 cells)
Mean force, 0–6 cells
Line charge $\pi^2 K\rho/2$
(d)
Escape vs restoring force
Escape probability p_{esc}
Peak elastic restoring force (k_BT/cell)
Labels: ρ
Circles
Aspect ≥ 1.4